

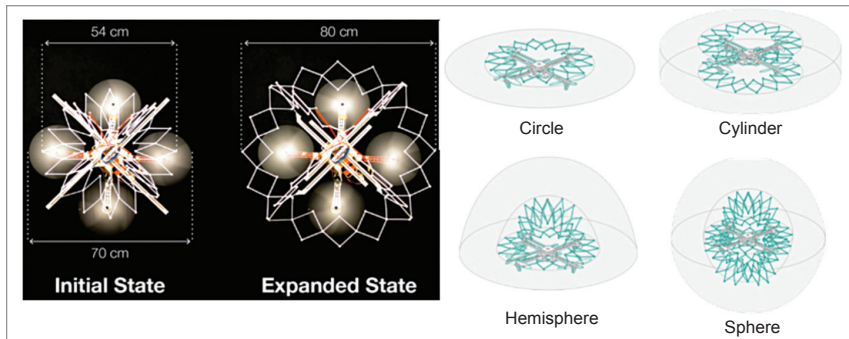


Fig. 6: Pufferbot and its various shapes (Courtesy: Hooman Hedayati, Ryo Suzuki, Daniel Leithinger, Daniel J. Szaflir)

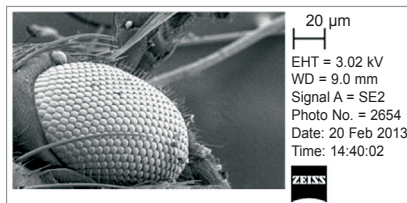


Fig. 7: Mosquito eye (Courtesy: WCFTO Nano SEM pictures)

The idea of increasing size by the pufferfish and the concept of vehicle airbags that inflate during crashes have been employed in Pufferbot (see Fig. 4 through Fig. 6). Pufferbot is teamwork of researchers from two universities—Atlas Institute, University of Colorado, and University of Calgary. It is an aerial robotic drone that expands to protect the propellers of the drone when it is close to obstacles, or when the robot is falling on the ground, or when it is navigating through cluttered spaces. The plastic hoops on top of its DJI Flame Wheel F450 frame acts as the shield and prevents unnecessary contact with the surrounding environment. The rack and pinion mechanism with one degree of freedom actuation is used to actuate the hoops.

The team used a proportional integral derivative (PID) controller to maintain the position and altitude of the aerial drone. The PID helps control the drone and its expandable structure separately by a central computer or by a teleoperator. Academy Award winner Vicon motion capture system is embedded in the device for tracking and for wireless communication to inform the microcontroller to control the size of the expandable structure. If the device is manually handled, the teleoperator takes charge of controlling the robot and the expandable structure. One important virtue of this

expandable structure design is, it can vary its shape depending upon the surrounding environment and make better navigation.

Insect inspired hydrophobia

In the design of engineering structures, the microstructure of materials is an essential feature because micro-characterisation helps build fatigue strength and increases life cycle of the material. ‘Lotus Effect’ was discovered

in 1970 by German botanist Prof. Dr Wilhelm Barthlott, who used nanotechnology to mimick the property in artificial surfaces to repel liquid, which finds useful applications in scientific, industrial, and practical platforms. Artificial liquid-repellent surfaces are significant for self-cleaning, anti-icing, and anti-reflection technologies, water harvesting, and droplet manipulation.

Apart from following this traditional method, Pennsylvania State University researchers, Lin Wang *et al*, garnered information about nanoscopic hairs on insects. The nanoscale features on insects’ skin give them the essential property for survival.

For example, the eye of a mosquito (Fig. 7 and Fig. 8) is surrounded by densely packed nanoscopic hairs which are referred to as ‘solid fraction textures’ in engineering. Solid fraction textures of insects range from 100nm to 300nm, which adds more water-repelling benefits than the lotus effect

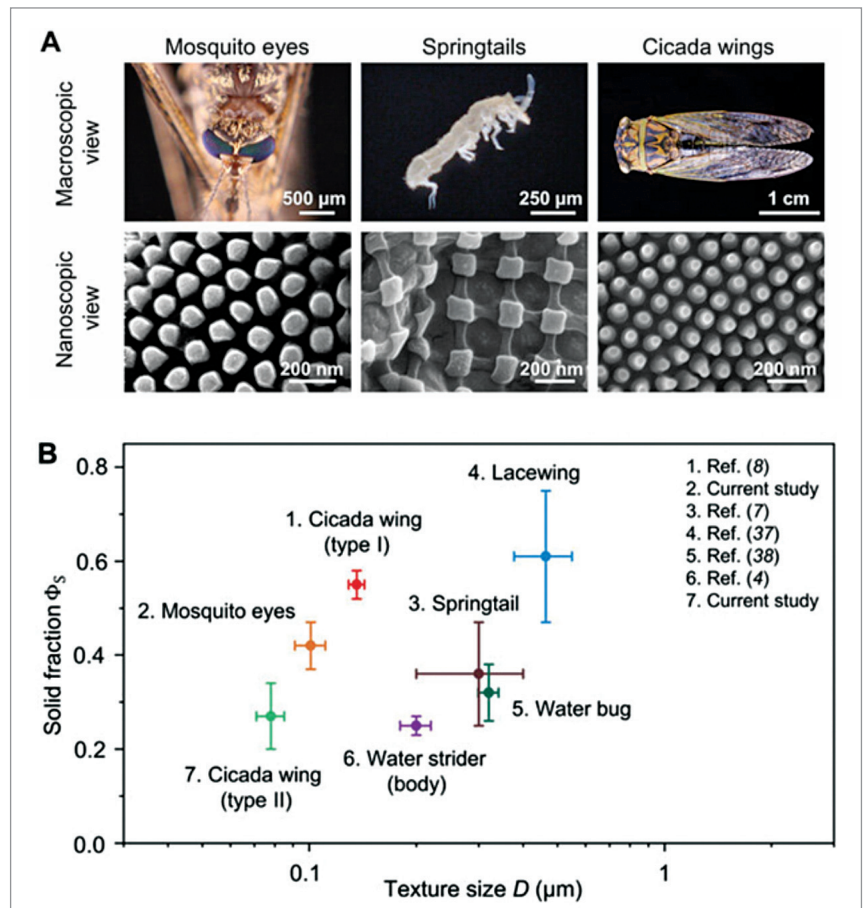


Fig. 8: Water-repellent insects with high solid fraction nanoscale surface texture (Courtesy: Lin Wang, Ruoxi Wang, Jing Wang, Tak-Sing Wong, Pennsylvania State University)